

Astaroth Elizabethson, Sc.D.
Bogotá D.C., Colombia
(+57) 3124229194
lizon169@gmail.com



PROFESSIONAL PROFILE

Physicist and mathematician with a passion for teaching. With experience in education and programming (Mathlab, Java, R and SQL), specialized in supervised machine learning applied to astronomic databases. Possesses strong social skills that enable effective teaching and leadership, alongside a proven ability to work well within a team.

EXPERIENCE

BACU, Bogotá D.C., Colombia. **Oct 2024- Present**
Operational Control and Data Support Analyst, duties include ensuring the consistency of the required data within the company, maintaining the quality and accuracy of data across all company systems, performing data analysis to identify trends, patterns, and opportunities for improvement, and providing reports and recommendations based on data analysis to support decision-making. Data is managed using SQL (postgreSQL and MySQL), while Grafana is used as the reporting interface tool.

Universidad de los Andes, Bogotá D.C., Colombia. **2015-2016**
Graduate Teaching Assistant, Department of Physics: responsibilities included delivering supplementary or experimental classes on fundamental physics topics (classical mechanics, thermodynamics, electromagnetism) to first-semester university students majoring in engineering and natural sciences.

Positions held:

- Graduate Teaching Assistant for Master's Program (2015-2016, four academic semesters).
- Lecturer (inter-semester period 2015).
- Research Assistant (inter-semester period 2016).
- Graduate Teaching Assistant for Doctoral Program (2017- April 2023).
- Lecturer (inter-semester, second semester 2023).

Ingenius Colombia, Bogotá D.C., Colombia. **2013–2014**
Taught classes and provided personalized tutoring to students from different universities in topics such as basic kinematics, differential, integral, and vector calculus, linear algebra, and differential equations (<https://ingeniuscolombia.com>).

Universidad de los Andes, Bogotá D.C., Colombia. **2010-2014**

- **Teaching Assistantships:** Involved assisting the professor in grading assignments and exams, as well as providing supplementary guidance to students in Vector Calculus, Integral Calculus, Linear Algebra, Basic Physics, and Probability.
- **Teaching Practicums:** Entailed delivering supplementary classes to groups averaging 25 bachelor students, focusing on practical applications of theoretical concepts covered in lectures. Classes: Linear Algebra, Integral Calculus, and Physics II: Basic concepts of electromagnetism and thermodynamics.

EDUCATION

Doctor of Science in Physics. Universidad de los Andes, Bogotá D.C., Colombia. 2017-2023

GPA: 4.40/5.0

Relevant Courses, electives complementing the mandatory curriculum:

- Non-Parametric Statistics: Introduction to the properties and applications of non-parametric estimators in statistical algorithms. Bootstrapping and bagging methods.
- Pattern Recognition: Supervised and unsupervised learning methods, along with their mathematical background.
- Machine Learning: Practical course on various supervised learning methods, their programming, and application to different problems.

Thesis: Variable stars' light curve detection and classification using supervised machine learning. (<https://hdl.handle.net/1992/73980>).

Master of Science in Physics. Universidad de los Andes, Bogotá D.C., Colombia. 2015-2016

GPA: 4.25/5.0

Relevant Courses, electives complementing the mandatory curriculum:

- Open Clusters: Photometric processing (via IRAF) in sparse fields.
- Advanced Astrophysical Methods: Processing of spectroscopic data.

Thesis: *Supervised Techniques Applied to Selected Fields of the VVV Survey*. (<http://hdl.handle.net/1992/13966>).

Bachelor of Science in Mathematics. Universidad de los Andes, Bogotá D.C., Colombia. 2008-2014

GPA: 3.86/5.0

Relevant Courses, electives complementing the mandatory curriculum: Dynamic Systems, Game Theory, and Number Theory.

Bachelor of Science in Physics. Universidad de los Andes, Bogotá D.C., Colombia. 2008-2014

GPA: 3.86/5.0

Relevant Courses, electives complementing the mandatory curriculum: Astrophysical Methods (Experience in IRAF program, processing astronomical images obtained with CCDs), Statistical Mechanics, Advanced Quantum Physics.

PUBLICATIONS

Elizabethson, A., Serna, J., García-Varela, A., Hernández, J., & Cabrera-García, J. F. (2023). Machine-learning Morphological Classification of TESS Light Curves of T Tauri Stars. *The Astronomical Journal*, 166(5), 189. DOI: [10.3847/1538-3881/acf865](https://doi.org/10.3847/1538-3881/acf865)

Elizabethson, A., García-Varela, A., Pérez-Ortiz, M. F., Sabogal, B. E., & Minniti, D. (2017, July). An Application of Supervised Learning Methods to Search for Variable Stars in a Selected Field of the VVV Survey. In *Revista Mexicana de Astronomía y Astrofísica Conference Series (Vol. 49, pp. 97-97)*. Bibcode: [2017RMxAC..49...97R](https://ui.adsabs.org/abs/2017RMxAC..49...97R)

LANGUAGES

Spanish:	Native.
English:	B2.